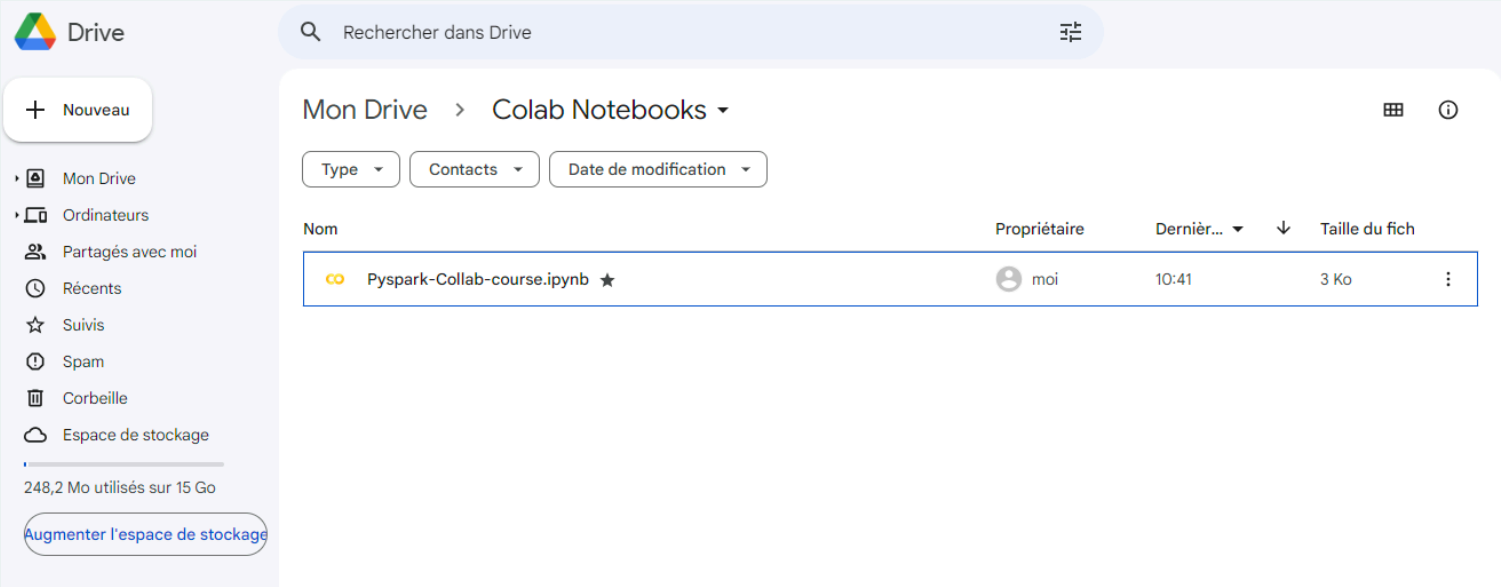
# Hadoop & Spark - ISEN 2025/2026

## Lab: Setup PySpark on Google Collab

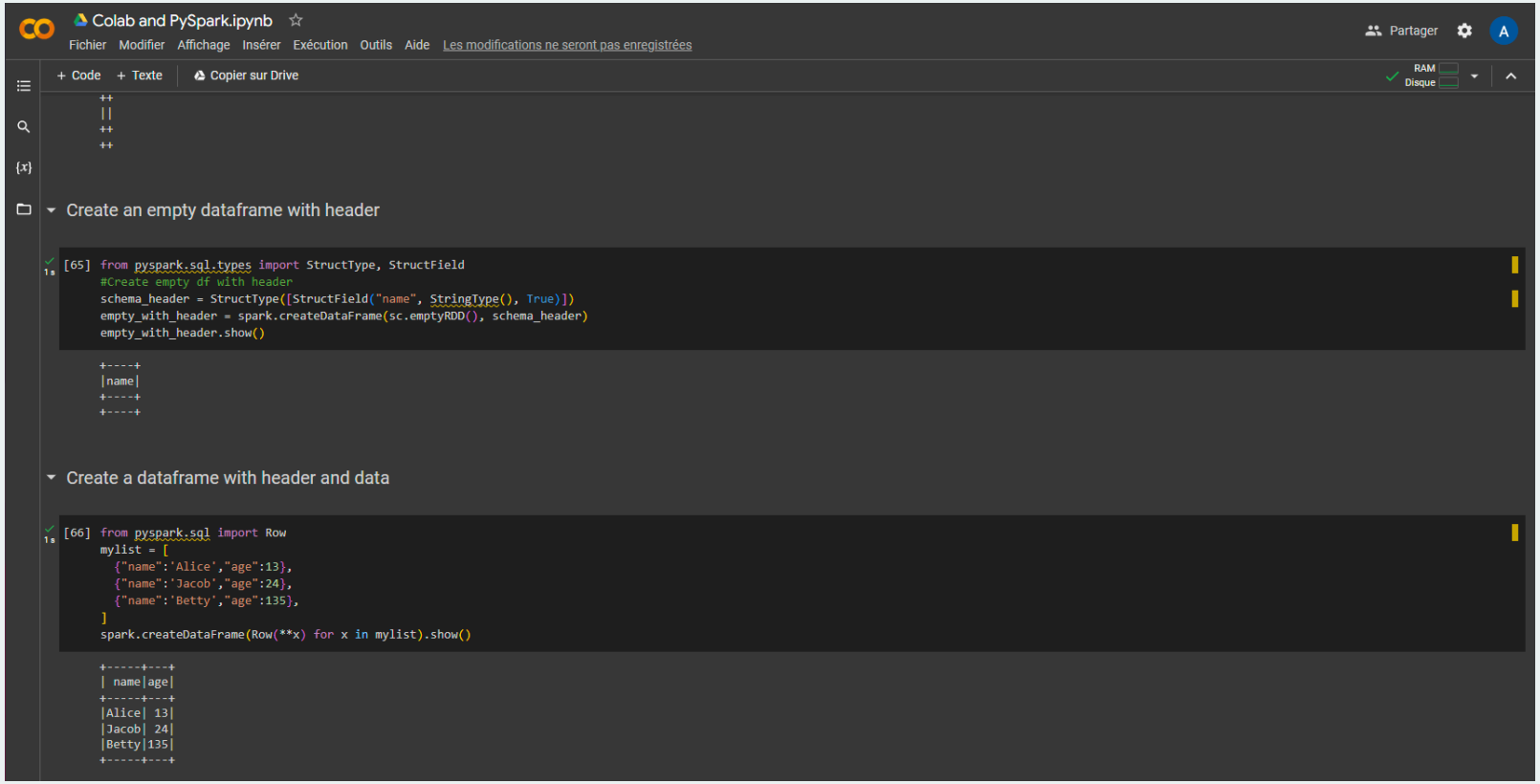
# For this Lab you will need:

- Ideally, prior Python knowledge
- A free Google Account (For Google Drive and Google Collab)
- Space left on your Google Drive Account (First 15Go are free)
- A free Kaggle Account
- A free NGROK Account



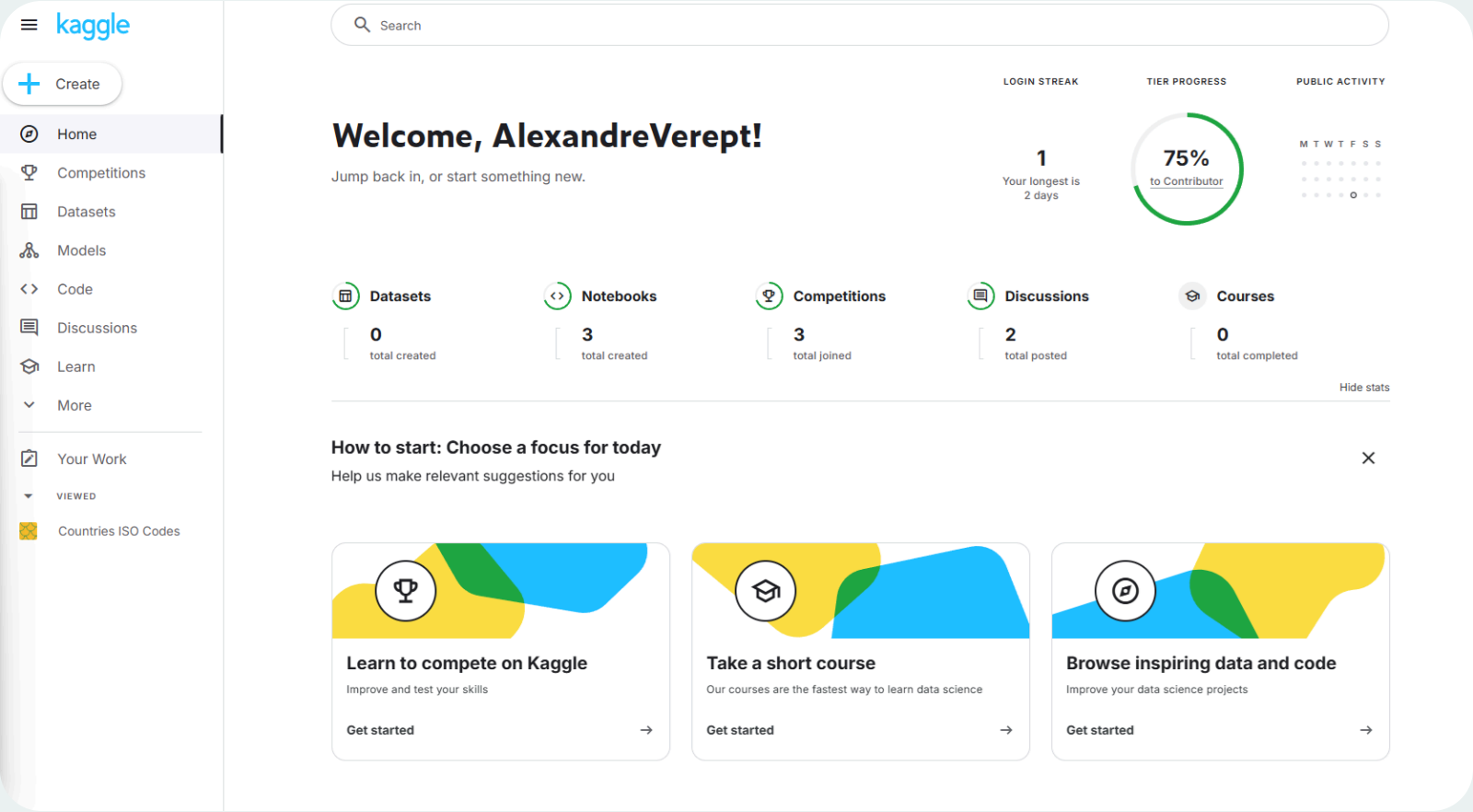
# What is Google Collab ?

Google Collab is a free tool provided by Google to work on Python Notebooks.



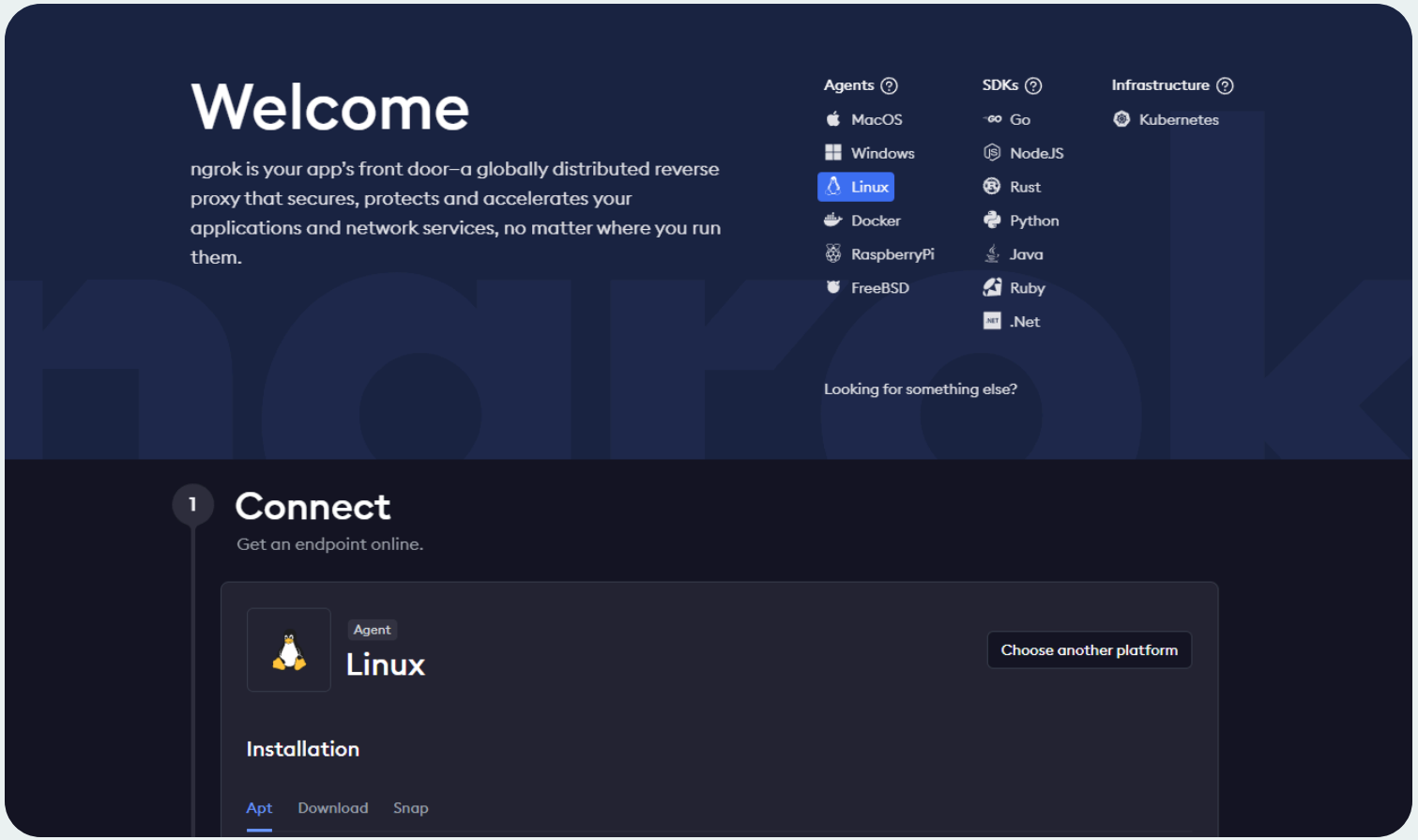
# What is Kaggle ?

Kaggle is a website were you can find free datasets and data competitions with a large datascience community.



# What is NGROK ?

NGROK is a unified ingress platform. This will allow us to redirect the local urls of the Google Collab Notebooks to a url we can access.



# Install PySpark for Google Collab (1/3)

Install Dependencies (may take up to 5 minutes in Collab) :

- Java 8
- Apache Spark with hadoop and
- Findspark (used to locate the spark in the system)

```
!apt-get install openjdk-8-jdk-headless -qq > /dev/null
!wget -q http://archive.apache.org/dist/spark/spark-3.5.8/spark-3.5.8-bin-hadoop3.tgz
!tar xf spark-3.5.8-bin-hadoop3.tgz
!pip install -q findspark
```

# Install PySpark for Google Collab (2/3)

Set environment variables:

```
import os
os.environ["JAVA_HOME"] = "/usr/lib/jvm/java-8-openjdk-amd64"
os.environ["SPARK_HOME"] = "/content/spark-3.5.8-bin-hadoop3"
```

Set and import findspark:

```
import findspark
findspark.init()
```

# Install PySpark for Google Collab (3/3)

Set and import Pyspark properties:

```
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("CollabApp").master("local[*]").config('spark.ui.port', '4050').getOrCreate()
spark.conf.set("spark.sql.repl.eagerEval.enabled", True)
```

And finally launch Spark:

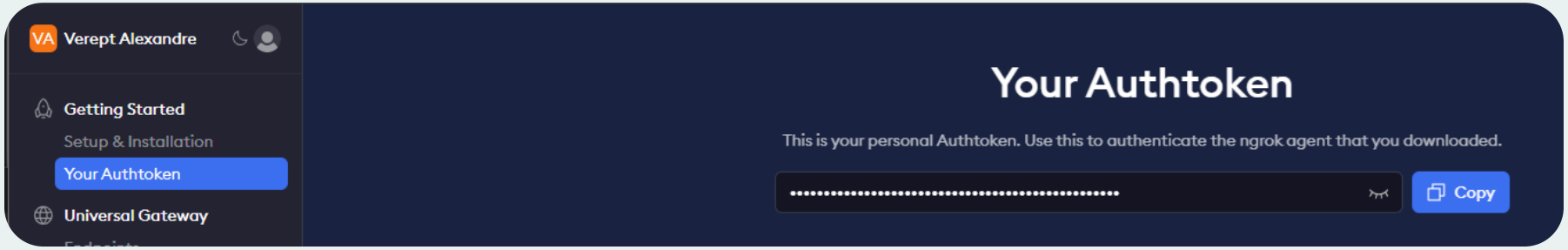
```
spark
```

# Show Spark UI using ngrok (1/2)

**Important**: you need to create a NGROK account: https://ngrok.com/

You will need to register, retrieve your token on the website and verify your email address **before** launching the code.

You will find the auth token once logged in:



# Show Spark UI using ngrok (2/2)

Install ngrok:

```
!curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc \
        | sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null \
        && echo "deb https://ngrok-agent.s3.amazonaws.com buster main" \
        | sudo tee /etc/apt/sources.list.d/ngrok.list \
        && sudo apt update \
        && sudo apt install ngrok
```

Set your authentication token :

```
get_ipython().system_raw('ngrok config add-authtoken Your-token-here') # <- change here
get_ipython().system_raw('ngrok http http://localhost:4050 &')
```

Get the url here:

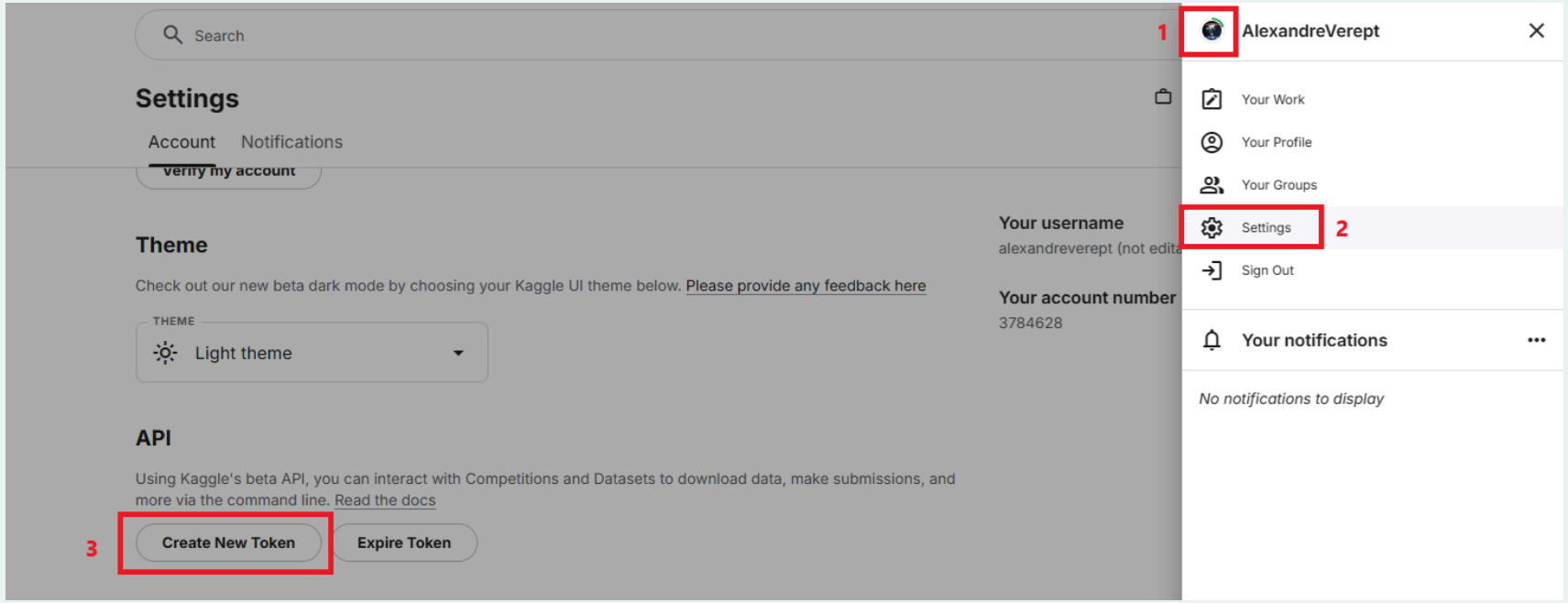
```
!curl -s http://localhost:4040/api/tunnels | grep -Po 'public_url":"(?=https)\\K[^"]*'
```

# Download & load a dataframe from Kaggle (1/4)

For this section to work, you have to first create a Kaggle account and copy you API token.

Kaggle

2026 -> Go to Legacy token instead 🚩



# Download & load a dataframe from Kaggle (2/4)

Install the kaggle python module:

```
! pip install -q kaggle
```

Upload you kaggle.json downloaded in the previous slide (you can drag and drop):

```
from google.colab import files
files.upload()
```

Create a folder with the token and give the permissions needed:

```
!mkdir ~/.kaggle
!cp kaggle.json ~/.kaggle/
!ls ~/.kaggle
! chmod 600 ~/.kaggle/kaggle.json
```

# Download & load a dataframe from Kaggle (3/4)

And you can download any dataset with this command:

```
!kaggle datasets download -d jkraak/lego-sets-and-themes-database
```

(link to the example dataset)

You may have to unzip it:

```
!unzip lego-sets-and-themes-database.zip
```

# Download & load a dataframe from Kaggle (4/4)

With Spark imported previously, you only have to read it now:

```
df = spark.read.csv('lego_sets_and_themes.csv', header=True, sep=",")
df.show(5)
```

```
df = spark.read.csv('lego_sets_and_themes.csv', header=True, sep=",")
df.show(5)

+----------+--------------------+------------+--------------+--------------------+----------+
|set_number|            set_name|year_released|number_of_parts|           image_url|theme_name|
+----------+--------------------+------------+--------------+--------------------+----------+
|     001-1|               Gears|       1965.0|          43.0|https://cdn.rebri...|   Technic|
|     002-1|4.5V Samsonite Ge...|       1965.0|           3.0|https://cdn.rebri...|   Technic|
|    1030-1|TECHNIC I: Simple...|       1985.0|         210.0|https://cdn.rebri...|   Technic|
|    1038-1|  ERBIE the Robo-Car|       1985.0|         120.0|https://cdn.rebri...|   Technic|
|    1039-1|Manual Control Set 1|       1986.0|          39.0|https://cdn.rebri...|   Technic|
+----------+--------------------+------------+--------------+--------------------+----------+
only showing top 5 rows
```

**Note:** PySpark commands are often very similar to pandas

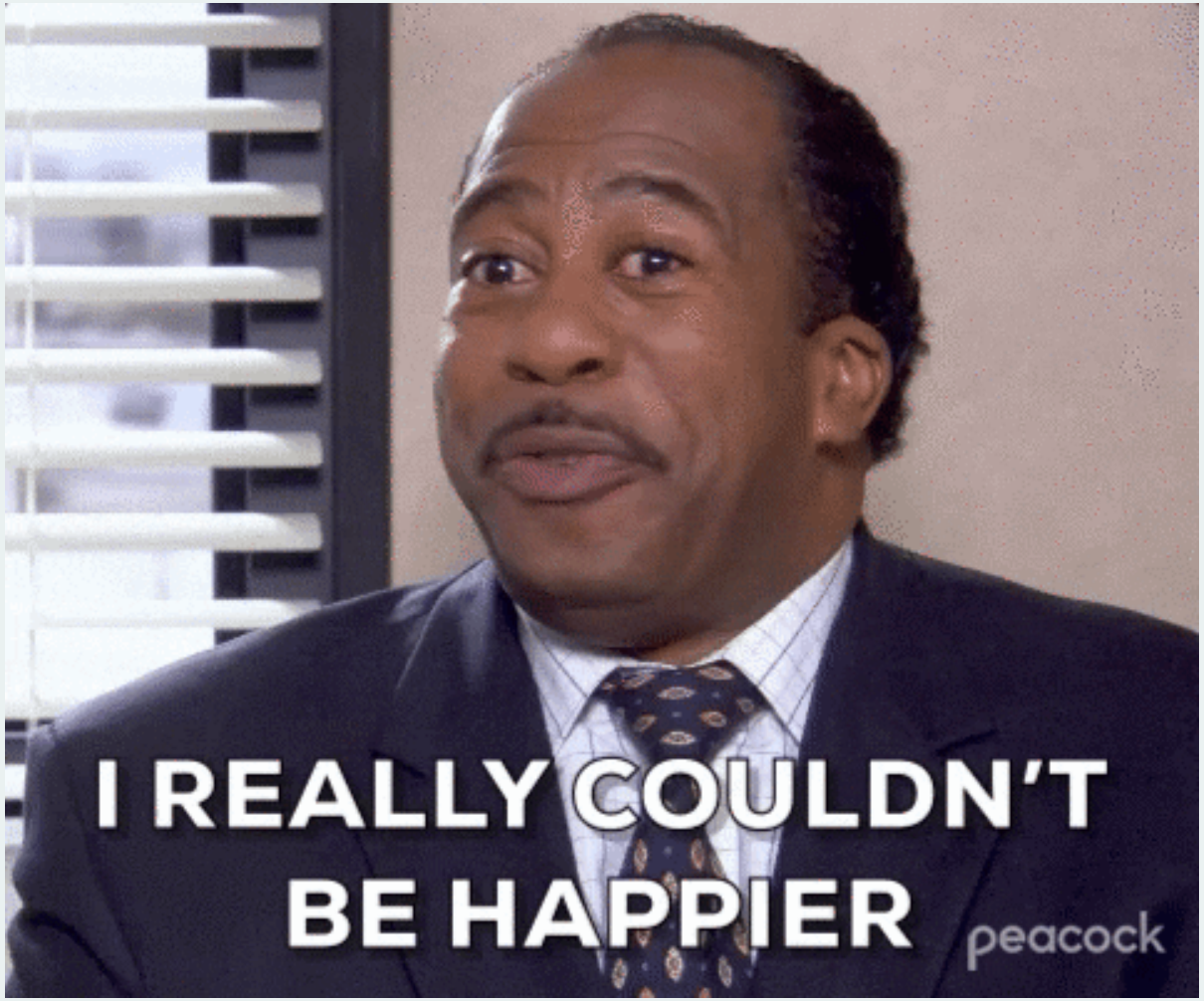
# Prepare for later

We cast 2 columns and set the delimiter as ';' to make our life easier later on (trust me):

```
from pyspark.sql.types import *

df = df.withColumn("number_of_parts", df["number_of_parts"].cast(IntegerType()))
df = df.withColumn("year_released", df["year_released"].cast(IntegerType()))

df.write.options(delimiter=';').mode("overwrite").csv("lego_sets_and_themes_clean.csv")
```

# Congrats

You are now ready to use Spark on your Google Collab environment 🎉